



Week Playbook — Senior Engineers • Ages 14-17

Owning Your Superpower: one week from gamer → game developer → leader, coding a real platformer in **Godot**. This is YOUR track at a glance — every workshop, lab, and link, colour-coded like the rest of the kit.

What you'll accomplish

Senior Engineers • 14-17

Code a fully working **2D platformer** from scratch in Godot — the professional lifecycle.

The Ultimate Challenge

Everyone: turn your own original idea into a live, launch-ready platformer.

Team Superhero Comic

In groups of 3–4: write, design & publish a superhero comic with real world-building.

🔑 **Endless platform (14-17):** [sign in here](#) with access code **BGC26IEGO** (facilitators: **BGC26IEGOAdmin**)

Your digital toolbelt

TOOL	WHAT IT'S FOR
Canva	Your creative studio — comic books, moodboards, storyboards.
Threadbare	Interactive intro to core game mechanics (10–13 track).
Godot Engine	The professional game engine where you write real code.
Piskel	Pixel-art editor for sprites, tiles & frame-by-frame animation.
Google Gemini	AI brainstorming partner for concept art, lore & creative text.
Asset Folder Library	Ready-to-use characters, animations & tilemap worlds for your game.
itch.io free assets	Even more downloadable art if you want it.

Your milestones (how you win the week)

1. **Technical Mastery** — finish the full 10-session Godot curriculum.
2. **Collaborative Leadership** — publish your group's Superhero Comic in Canva, weaving in the morning leadership frameworks.
3. **Advanced Microcredential** (*optional*) — complete 5 extra Threadbare cross-training sessions and earn an Arizona State University micro credential (15 total lessons).
4. **Creative Innovation Challenge** (*optional*) — custom mechanics, secret levels, or your own scoring system.

Every day's rhythm

9:30 - 9:55

 Kickoff & Breakfast

9:55 - 10:00

 The Daily Brain-Teaser

10:00 - 10:30

 Empowerment Hour — Girls Leadership Warm-Up

10:30 - 12:00

 Deep Dive — Morning Coding Lab

12:00 - 1:00

 Fuel Break (Lunch)

1:00 - 3:15

 Creator Mode — Afternoon Coding Session

3:15 - 3:45

 Team Challenge — Superhero Comic Design

3:45 - 4:00

 Mission Debrief & Sign-Out

The five days

MON

The Design Process & Redefining Leadership



Questions of the day

Imagine leadership as a personal superpower — what does it look like when it lifts others up? · How does coding bridge a wild imagination to a finished, working project?

LEADERSHIP Workshop 1 · Redefining Leadership

Forget "loudest person in the room." Real leadership = **empathy, active listening, confidence, community impact**. Find how your traits make you a natural leader.

- [Lean In Girls guide](#) → Ignite Session (pg. 7–9) · Vision Board (pg. 10–13)

CODING Morning Lab · Game Mechanics & Deconstruction

Great engineers observe first. Watch [this game being played](#) and break it into: the **Mechanics** (rules), the **Technical Blueprint** (code & nodes behind it), and the **Narrative Atmosphere** (story & biomes).

- **14–17:** Godot Sessions 1 & 2 (Endless link + code above)

CODING Afternoon Lab · Building Custom Character Assets

- **14–17:** Godot Sessions 3 & 4

YOUR GAME Create-Your-Own-Game track · today's videos

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|--|-----------|
|  How to make a moodboard in Canva | Recipe #2 |
|  Moodboard tutorial (Loom) | Recipe #2 |
|  2D tilesets with Godot & Piskel | |
|  Opening Godot & creating a project | Recipe #3 |
|  Import game assets into Godot | Recipe #4 |
|  Creating a Player in Godot | Recipe #5 |
|  Add Player scene & playtest | Recipe #6 |

COMIC Leadership Lab 1 · Launch your Superhero Comic

Teams of 3–4. Turn your personal "superpowers" into superhero characters, and map the intro with the **Hero's Journey Circle** (the 12-step roadmap movie writers use).

- [What is the Hero's Journey?](#) · [Comic template in Canva](#) (**Recipe #1**)

TUE Gameplay Mechanics & Unlocking Superpowers

Question of the day

If you could unlock any superpower today, what would it be — and how are your real-life best qualities already your real-world superpowers?

LEADERSHIP Workshop 2 · Discovering Your Leadership Superpowers

Empathy, organization, artistic vision, problem-solving, humor — all legitimate leadership superpowers. Map your top strengths and how they help your team.

- [Leadership guide](#) → Strengths Exploration (pg. 37–40)

CODING Morning Lab · Core Gameplay & Programming Systems

A game without mechanics is a static picture. Today you write the rules of your universe.

- **14–17:** Godot Sessions 5 & 6

CODING Afternoon Lab · Interactive Loops & UX Design

- **14–17:** Godot Sessions 7 & 8

YOUR GAME Create-Your-Own-Game track · today's videos

- | | |
|---|------------|
|  Adding a script, camera & boundary | Recipe #7 |
|  Changing player speed with constants | Recipe #8 |
|  Tilemaps & platforms | Recipe #9 |
|  Painting a background with tilesets | Recipe #10 |
|  Adding platform scenes & collisions | Recipe #11 |
|  Animating moving platforms with keyframes | Recipe #12 |
|  Key-binding movement & jump animations | Recipe #13 |
|  Fix player animations with code | Recipe #14 |

COMIC Leadership Lab 2 · Superpowers into your script

Advance through **Stages 4–8** of the Hero's Journey: write this morning's leadership superpowers into your characters' dialogue and choices, cross the threshold, meet allies, face the first big tests.

WED User Experience & Challenging Stereotypes



Questions of the day

What is a "stereotype," and how do they form? · Can a stereotype ever be truly "positive," or are they naturally limiting?

LEADERSHIP Workshop 3 · Challenging Perspectives & Breaking Barriers

How stereotypes hold people back — especially in tech — and how to spot them, break barriers, and build a team where everyone's authentic identity counts.

- [Leadership guide](#) → Unpacking Stereotypes (pg. 11–14) · Stereotypes Stomp (pg. 15–16)

CODING Morning Lab · Optimization & Progression

- **14–17:** Godot Sessions 9 & 10 — build a **HUD**, finish player physics. 🎉
Technical intensive complete!

YOUR GAME Afternoon · Independent Developer Time

Both cohorts graduate to **open studio**. Build your own levels: scoreboards & on-screen text, audio systems, and clean folders ready to export. Today's toys all have recipes in the SOP Book:



Restart zones

Recipe #15



Coins with player-only detection

Recipe #16



Patrolling enemies

Recipe #17



Fall effect on enemy hit

Recipe #18



On-screen instructions

Recipe #19



Jump & coin sound effects

Recipe #20



Score tracking & win message

Recipe #21

COMIC Leadership Lab 3 · Climax & Resolution

Push through **Steps 8–12:** the Supreme Ordeal (your heroes' combined superpowers vs. their greatest challenge), intentionally **defeat a real-world stereotype**, and Return with the Elixir — home, changed, ready to lift their community.

THU

Playtesting, Feedback & Presentation Prep



Question of the day

Recall a moment you felt nervous about a brand-new challenge. What inner strength pushed you through — and how does that courage make you a leader who inspires others?

LEADERSHIP

Workshop 4 · Inspiring Girls to Go For It

Ideas + skills need one more thing: the **courage to step into the spotlight**. Practice speaking confidently about your work and turning this week into real-world action.

CODING

Morning Lab · QA & Peer Playtesting

- **Hour 1 — The Polish:** final touches on terrain, sound levels, animations.
- **Hour 2 — The Exchange:** swap laptops! Hunt for bugs, broken collisions, unclear paths in each other's games.
- **The Feedback Loop:** fill the official *Playtest Feedback Form* — constructive, clear, kind.

CODING

Afternoon · Feedback → Polish + Slides

- **Rapid prototyping:** fix at least one major issue your playtester found.
- **Developer Spotlight slide:** your concept, code snippets, Piskel art, and growth story.
- **Superhero Team pitch deck:** lore, Canva art pages, and how the Hero's Journey showcases real leadership.

SHOWCASE

Evening · Dress Rehearsal (Science-Fair format)

Stations set: game loaded on one screen, slides on another, one device showing the team comic. Room splits into two shifts — Shift A presents while Shift B tours and asks questions, then swap after 20 minutes.

FRI

Exhibition Day & Career Pathways



Question of the day

How have you evolved since Monday morning? Which achievement — which newly unlocked superpower — makes you proudest?

CAREERS Morning · Mapping Tech Careers

Everything you did this week is a real, paid job somewhere:

CAREER	WHAT THEY DO	YOU DID IT WHEN YOU...
Gameplay Programmer	Core scripts, physics, collisions	wrote movement scripts & restart loops in Godot
Level Designer	Maps worlds, balances difficulty	placed tiles, spikes, coins & progression
Technical Artist	Characters & frame-by-frame animation	drew avatars, monsters & tiles in Piskel
Creative Director	Lore, worlds, dialogue	drafted profiles & scripts for your comic
Project Manager	Leads teams, keeps everyone unblocked	ran your group with Lean In frameworks

These exact pipelines are used daily at **Nintendo, Epic Games, Roblox, Pixar, Cartoon Network, Google, and Apple.**

SHOWCASE Afternoon · The Official Project Showcase 🎉

1. Hand guests the keyboard — let them **play your level**.
2. Walk them through a coding challenge you hit and **how you solved it**.
3. Present your team's comic — how your real superpowers guided your hero's journey.

(Complete your post-surveys first — they're in the student handout.)

NEXT STEPS Keep building your superpowers

- **Download Godot** — it's 100% free. Copy your project home and keep expanding it.
- **Virtual workshops:** wearebgc.org
- **Code-along episodes:** bit.ly/bgccodealong

Your creativity + leadership + code = endless possibilities. Keep building! 🐟

Polished from the BGC "Leading Through Code: Owning Your Superpower" playbook — same content, same links, kit colour-coding. **Recipe #N** tags point into the companion Game-Build SOP Book (the appendix, polished separately).